(FP1) 6. Vectors 专题作业

NO. 1

The position vectors of the points A, B, C, D are

$$\mathbf{i} + \mathbf{j} + 3\mathbf{k}, \quad 3\mathbf{i} + 4\mathbf{j} + 5\mathbf{k}, \quad -\mathbf{i} + 3\mathbf{k}, \quad m\mathbf{j} + 4\mathbf{k},$$

respectively, where m is a constant.

i. Show that the lines AB and CD are parallel when $m = \frac{3}{2}$. [1]

ii. Given that $m \neq \frac{3}{2}$, find the shortest distance between the lines AB and CD. [5]

iii. When $m = 2$, find the acute angle between the planes ABC and ABD, giving your answer in degrees. [6]

NO. 2

The plane Π_1 has equation

$$\mathbf{r} = \begin{pmatrix} 5 \\ 1 \\ 0 \end{pmatrix} + s \begin{pmatrix} -4 \\ 1 \\ 3 \end{pmatrix} + t \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix}.$$

i. Find a cartesian equation of Π_1. [3]

The plane Π_2 has equation $3x + y - z = 3$.

ii. Find the acute angle between Π_1 and Π_2, giving your answer in degrees. [2]

iii. Find an equation of the line of intersection of Π_1 and Π_2, giving your answer in the form $\mathbf{r} = \mathbf{a} + \lambda\mathbf{b}$. [5]

NO. 3

The plane Π_1 passes through the points (1, 2, 1) and (5, −2, 9) and is parallel to the vector $\mathbf{i} + 2\mathbf{j} + 3\mathbf{k}$.

i. Find the cartesian equation of Π_1. [4]

The plane Π_2 contains the lines

$$\mathbf{r} = 2\mathbf{i} - 3\mathbf{j} + \mathbf{k} + \lambda(\mathbf{i} - 2\mathbf{j} - \mathbf{k}) \text{ and } \mathbf{r} = 2\mathbf{i} - 3\mathbf{j} + \mathbf{k} + \mu(2\mathbf{i} + 3\mathbf{j} - \mathbf{k})$$

ii. Find the cartesian equation of Π_2. [4]

iii. Find the acute angle between Π_1 and Π_2. [3]

NO. 4

The plane Π_1 has equation r = –4j – 3k + λ(i – j + k) +μ(i + j – k).

a. Obtain an equation of Π_1 in the form $px + qy + rz = d$. [4]

b. The plane Π_2 has equation **r**.(–5**i** + 3**j** + 5**k**) = 4.
Find a vector equation of the line of intersection of Π_1 and Π_2 . [4]

The line l passes through the point A with position vector a**i** + a**j** + $(a - 7)$**k** and is parallel to (1–b)i + b**j**+b**k**, where a and b are positive constants.

c. Given that the perpendicular distance from A to Π_1 is $\sqrt{2}$, find the value of a. [2]

d. Given that the obtuse angle between l and Π_1 is $\frac{3}{4}\pi$, find the exact value of b. [4]

NO. 5

The position vectors of the points A, B, C, D are

$$7\mathbf{i} + 4\mathbf{j} - \mathbf{k}, \quad 11\mathbf{i} + 3\mathbf{j}, \quad 2\mathbf{i} + 6\mathbf{j} + 3\mathbf{k}, \quad 2\mathbf{i} + 7\mathbf{j} + \lambda\mathbf{k}$$

respectively.

a. Given that the shortest distance between the line AB and the line CD is 3, show that $\lambda^2 - 5\lambda + 4 = 0$. [7]

Let Π_1 be the plane ABD when $\lambda = 1$.
Let Π_2 be the plane ABD when $\lambda = 4$.

b. i. Write down an equation of Π_1, giving your answer in the form $\mathbf{r} = \mathbf{a} + s\mathbf{b} + t\mathbf{c}$. [2]

ii. Find an equation of Π_2, giving your answer in the form $ax + + by + cz = d$. [4]

c. Find the acute angle between Π_1 and Π_2. [5]

NO. 6

The lines l_1 and l_2 have equations $\mathbf{r} = -\mathbf{i} - 2\mathbf{j}+\mathbf{k}+s(2\mathbf{i}-3\mathbf{j})$ and $\mathbf{r} = 3\mathbf{i} - 2\mathbf{k} + t(3\mathbf{i}-\mathbf{j}+3\mathbf{k})$ respectively.

The plane Π_1 contains l_1 and the point P with position vector $-2\mathbf{i} - 2\mathbf{j}+4\mathbf{k}$.

a. Find an equation of Π_1, giving your answer in the form $\mathbf{r} = \mathbf{a}+\lambda\mathbf{b} + \mu\mathbf{c}$. [2]

The plane Π_2 contains l_2 and is parallel to l_1.

b. Find an equation of Π_2, giving your answer in the form $ax+by+cz = d$. [4]

c. Find the acute angle between Π_1 and Π_2. [5]

d. The point Q is such that $\overrightarrow{OQ} = -5\overrightarrow{OP}$.
Find the position vector of the foot of the perpendicular from the point Q to Π_2. [4]

NO. 7

Let t be a positive constant.

The line l_1 passes through the point with position vector $t\mathbf{i}+\mathbf{j}$ and is parallel to the vector $-2\mathbf{i}-\mathbf{j}$. The line l_2 passes through the point with position vector $\mathbf{j}+t\mathbf{k}$ and is parallel to the vector $-2\mathbf{j}+\mathbf{k}$.

It is given that the shortest distance between the lines l_1 and l_2 is $\sqrt{21}$.

a. Find the value of t. [5]

The plane Π_1 contains l_1 and is parallel to l_2.

b. Write down an equation of Π_1, giving your answer in the form $\mathbf{r} = \mathbf{a}+\lambda\mathbf{b} + \mu\mathbf{c}$. [1]

The plane Π_2 has Cartesian equation $5x - 6y + 7z = 0$.

c. Find the acute angle between l_2 and Π_2. [3]

d. Find the acute angle between Π_1 and Π_2. [3]